

2X2 matrix transpose

/*asks the user to input a 2×2 matrix and prints its transpose.*/

```
#include<stdio.h>
```

```
int main(){
```

```
    int i,j;
```

```
    int matrix[2][2];
```

```
    int transpose[2][2];
```

```
    printf("Enter elements of matrix : \n");
```

```
    for(i=0;i<2;i++){
```

```
        for(j=0;j<2;j++){
```

```
            printf("Enter element [%d][%d] = ",i,j);
```

```
            scanf("%d",&matrix[i][j]);
```

```
        }
```

```
    }
```

```
    //display the entered matrix
```

```
    printf("Entered matrix is : \n");
```

```
    for(i=0;i<2;i++){
```

```
        for(j=0;j<2;j++){
```

```
            printf("%d ",matrix[i][j]);
```

```
            if(j==1)
```

```
                printf("\n");
```

```
    transpose[j][i]=matrix[i][j];
```

```
    }
```

```

    }

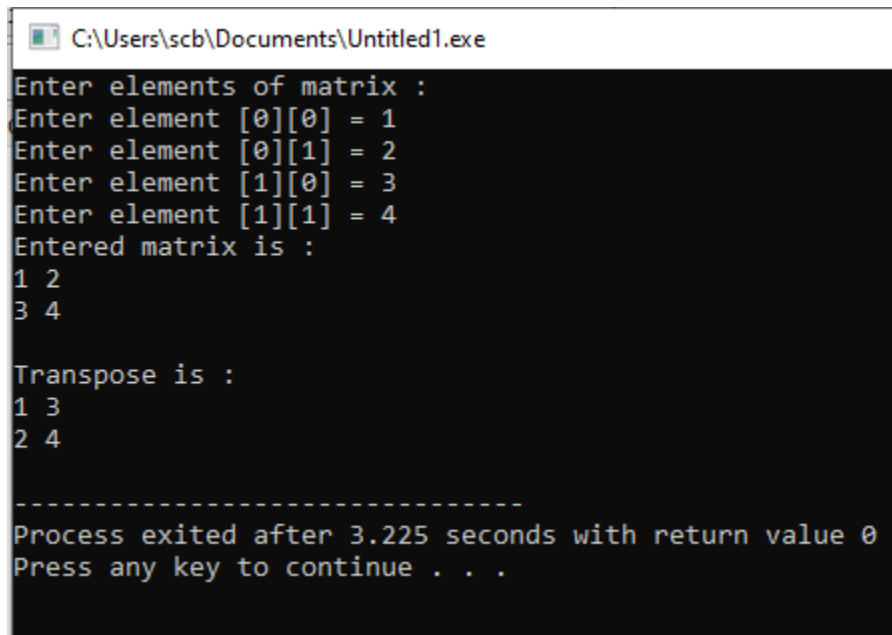
    printf("\nTranspose is : \n");

    for(i=0;i<2;i++){
        for(j=0;j<2;j++){
            printf("%d ",transpose[i][j]);

            if(j==1){
                printf("\n");
            }
        }
    }

    return 0;
}

```



```

C:\Users\scb\Documents\Untitled1.exe
Enter elements of matrix :
Enter element [0][0] = 1
Enter element [0][1] = 2
Enter element [1][0] = 3
Enter element [1][1] = 4
Entered matrix is :
1 2
3 4

Transpose is :
1 3
2 4

-----
Process exited after 3.225 seconds with return value 0
Press any key to continue . . .

```